

What is the price of a model’s eye?: Understanding How VLMs Answer Multiple Choice Questions in Multimodal Settings

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Abstract

MCQA is a key competence of vision-language models that is tested by a variety of mainstream benchmarks. However, these models can have a wide range of changes in performance when the task is diversified even slightly. The question we are asking is: *How do mainstream vision-language models perform on formatted multiple-choice question answering tasks?* We set out to evaluate and understand the models’ internal representations by employing methods like layer-wise activation patching, individual attention head localization, and vocabulary projection for answer choices to localize key hidden states and activation patterns that encode information about predictive tokens. We will also employ novel methods like visual-answer choices, comparison of symbol binding vs. visual grounding, spatial attention analysis, as well as cross-image activation patching so as to comprehensively probe the attention and activation information encodings within a model.

1 Introduction

Multi-choice question answering (MCQA) has been a popular task for evaluating LMs (Wiegrefe et al., 2025), and has been gaining traction in the space of vision-language models recently (Meng et al., 2024). A prominent example is the MMMU benchmark (Yue et al., 2024) which evaluates multimodal models on massive multi-discipline tasks on 28 open-source LLMs as well as proprietary models such as GPT-4V (Vision) and Gemini. While LMs have been evaluated comprehensively in this domain, VLMs are yet to be probed in depth.

Our methodology is rooted in NLP interpretability. Once an image is encoded and projected into the language model’s embedding space, it becomes a sequence of tokens that is treated exactly like text tokens. This allows us to probe the model’s language component and treat the vision encoder as a black box. Vocabulary projection and activation patching are defined in terms of a model’s

vocabulary and residual stream. Our vision specific experiments, spatial attention analysis, cross image activation patching and image as option binding, described later, all operate on image tokens in the language model’s residual stream, no intervention on the vision encoder itself happens.

Our project extends (Wiegrefe et al., 2025), an NLP interpretability paper on text-only language models. Our focus is not on whether the model answers correctly but on which layer and which attention heads produce that answer. While benchmarks like MMLU and MMMU are widely trusted, prior work shows that text-only models can be sensitive to superficial changes, such as answer order or label symbols (Pezeshkpour and Hruschka, 2024; Alzahrani et al., 2024; Khatun and Brown, 2024; Zheng et al., 2024). Our work falls within the line of this prior interpretability work and by adding vision as a new variable we ask whether this underlying brittleness extends to multimodal benchmarks.

2 Motivation

The use of vision language models is increasing in real world environments with their capabilities being judged predominantly from benchmarks featuring formatted multi-choice question answering (MCQA) such as MMMU. While these benchmark accuracy metrics provide a broad view of model’s abilities, they treat model as a black box and fail to reveal how the model arrives at a specific answer.

This presents a significant reliability risk as recent mechanistic interpretability work on text-only language models demonstrates that models often rely on superficial heuristics such as positional bias, answer order or label symbols, rather than semantic reasoning, while getting high benchmark scores. Modern VLMs are built by using these language models as a backbone hence the question of whether VLMs ground their answers in visual

evidence or do they utilise text-biased heuristics of their backbone models becomes important. A model could correctly identify an object in an image but map it to the wrong symbol answer due to inherited text processing bias and record a failure in the benchmark, or conversely, a model might guess correctly based on text symbols bias without utilizing visual input.

To ensure VLMs can be trusted in high-stakes tasks, understanding precise layers and attention mechanisms that govern multimodal MCQA becomes essential. By extending existing NLP interpretability techniques into multimodal space, we aim to understand whether model’s answers are driven by visual representations or its pre-existing textual heuristics

3 Contributions

Recent work by Wiegrefe et al. (Wiegrefe et al., 2025) studies the internal mechanisms underlying formatted MCQA in text-only language models. Using techniques such as vocabulary projection and activation patching, they identify layers and attention mechanisms involved in promoting specific answer choices, and further localize sparse attention heads responsible for answer-symbol prediction. We modify and extend this line of analysis to vision-language models, asking whether these mechanisms persist when answer selection additionally depends on visual information, and how visual information is incorporated into the answer-selection process.

Concretely, our contributions are: (i) testing whether the layer-wise and attention-head mechanisms Wiegrefe et al. identify for text-only MCQA persist across four VLM families; (ii) new diagnostics – answer-content vs. answer-symbol localization, and position vs. content disentanglement – for separating what a VLM represents internally from how that representation is expressed as a choice symbol; and (iii), time permitting, vision-specific extensions with no text-only analog, including image-as-option symbol binding and cross-image activation patching, which isolate the exact layer at which visual evidence enters the answer-selection pathway.

4 Methodology

In this project, we set out to evaluate and understand how models promote the prediction of specific answer choices, in the space of attention,

MLP, and different layers themselves. We plan to evaluate the models from families such as Qwen (Bai et al., 2025), PaliGemma (Beyer et al., 2024), LLaVA-OneVision (Li et al., 2024), and InternVL (Wang et al., 2025). Our proposed methods can be summarized as follows:

4.1 Core Methods

1. **Layer-wise Activation Patching:** Patch hidden states at the final token position from one prompt format (e.g., answer at position A) into another (answer at position B), layer by layer. Identifies which layers causally encode the answer choice in VLMs, and whether the critical layers are the same as in text-only LMs.
2. **Attention vs. MLP Decomposition:** Project MHSA and MLP outputs separately into vocabulary space at each layer. Determines whether attention heads still dominate discriminative answer symbol production over MLPs in VLMs, or if the MLP plays a bigger role due to visual feature integration.
3. **Individual Attention Head Localization:** Tests whether VLM answer production is still driven by a sparse set of attention heads or requires a broader set.
4. **Vocabulary/Logit Projection for Answer Choices:** Apply logit lens layer-by-layer: project each layer’s hidden state to the vocabulary and track how the logits for A/B/C/D evolve across depth. Shows the trajectory of answer symbol emergence in VLMs and whether it mirrors the smooth late-layer rise seen in text-only models.
5. **Answer-Content vs. Answer-Symbol Localization:** Separately track when the model’s hidden states encode the semantic content of the correct answer (e.g., "yellow") vs. the symbol (e.g., "A"). Reveals whether VLMs separate the "what is the answer" and "what letter is it" computations into distinct layers like text LMs do.
6. **Position Information vs. Answer-Content Information:** Patch between prompts where the correct answer content stays the same but its position changes ($A \rightarrow C$). Disentangles which layers encode where the correct answer sits in the list vs. what the correct answer

actually is, and whether visual grounding adds an extra processing stage.

7. **Arbitrary Answer Symbols:** Replace A/B/C/D with out-of-distribution symbols (Q/Z/R/X, 1/2/3/4) and track logit trajectories. Tests whether VLMs exhibit the same two-stage process (first promoting A/B/C/D, then switching to OOD symbols at late layers) observed in text-only models.

4.2 Stretch Goals

We also intend to do these novel vision-specific experiments if the duration of the course project permits, otherwise we'll continue evaluation on our own time expense:

1. **Visual Answer Choices (Image-as-Option):** Present answer choices as images instead of text (e.g., colored patches, diagrams). The model must bind a visual option to a symbol letter (no string matching possible). Tests whether the symbol binding circuit works at all when answer content is non-textual, and whether it requires more layers or different heads.
2. **Cross-Image Activation Patching:** Keep text completely fixed, but patch in image token activations from a different image (where a different answer is correct). Track at which layer the answer symbol flips. This has no text-only analog; it cleanly isolates the exact layer where visual evidence enters the symbol selection pathway, since any change in the predicted symbol is purely caused by the patched image representations.
3. **Spatial Attention Analysis (Question-Driven vs. Option-Driven):** For questions that reference specific image regions ("What is in the top-left?"), track which image token positions the answer-critical attention heads attend to. Does the model attend to the region the question asks about, or does it attend to regions corresponding to each answer option and compare? Reveals whether visual reasoning is question-guided or option-guided.
4. **Failure Mode Decomposition (Symbol Binding vs. Visual Grounding):** Compare layer-wise logit projection trajectories for correct predictions vs. wrong predictions. For a wrong prediction, determine whether the

model first represents the wrong answer content and then correctly maps it to its corresponding option symbol $\rightarrow$ visual grounding failed but symbol binding is intact. If the correct answer content is represented but the model maps it to the wrong option symbol $\rightarrow$ symbol binding failed despite correct visual grounding. Separates upstream visual understanding failures from downstream answer-symbol binding failures.

5 Dataset

We evaluate on the MMMU benchmark (Yue et al., 2024), which spans massive multi-discipline tasks across text and visual reasoning and has been used to evaluate both open-source and proprietary vision-language models. MMMU items provide the base pool from which we construct, for each question, the four position variants and the alternate-symbol variants (Q/Z/R/X, 1/2/3/4) used in the per-model-family screening procedure and in the layer-wise activation patching experiments described above.

6 Evaluation Plan

Before applying any interpretability method, we first screen each model family for the model that answers formatted MCQA consistently, following (Wiegrefe et al., 2025): for each item we construct four versions with the correct answer at each of the four positions, plus versions using unusual answer symbols (Q/Z/R/X, 1/2/3/4), and take the minimum accuracy across all of these variants. This penalizes a model whose apparent accuracy comes from a positional or symbol-frequency bias rather than genuine answer selection, and we keep only the strongest model per family, requiring at least 20% above random accuracy, before proceeding. For activation patching, we only compute effects on instances a model predicts correctly under both prompt versions being compared, so that causal effects are not confounded with cases the model never understood in the first place.

At each layer, we report two complementary metrics: logit values (the score assigned to each answer symbol, the gap between the top and second-choice symbol, and the largest score assigned to any other vocabulary token as a baseline) and probits (softmax-normalized probabilities). Logits can reveal a component actively suppressing an answer choice, an effect that can vanish after normalization, while probits let us compare confidence fairly

across our four model families, which have different raw logit scales.

7 Expected Outcomes

We aim to unravel the layers and attention/MLP components responsible for answer selection in VLMs and determine how these mechanisms compare across model families. We will characterize whether models separately represent answer content and answer symbols, and assess their sensitivity to answer position and symbol choice. The project shall also yield mechanistic evidence for whether visual information is integrated into the answer-selection pathway and, through the additionally proposed vision-specific experiments, distinguish failures of visual grounding from symbol binding, if time permits.

8 Timeline

Dates	Milestone
Aug 28 – Sep 7	Set up all four models and build the question-variant generator.
Sep 8 – Sep 21	Screen models and select the best one per family; build the activation-patching pipeline; begin layer patching and logit-lens analysis.
Sep 22 – Sep 30	Complete layer patching, attention/MLP decomposition, and logit-lens analysis on at least two model families.
<i>Mid-submission: September 30</i>	
Oct 1 – Oct 12	Complete the remaining core experiments across all four model families.
Oct 13 – Oct 24	Conduct stretch experiments, prioritizing cross-image patching and failure-mode analysis, followed by remaining experiments if time permits.
Oct 25 – Oct 31	Consolidate results, prepare figures, and write the final report.
<i>Final submission: October 31</i>	

Table 1: Project Timeline

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